A Surrogate Model for Reducing the Computational Cost of Neuron Simulations

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Introduction

- Multi-compartment models represent the spatial morphology of a neuron as connected compartments, reproducing neuronal activity in fine detail.
- Each compartment is described by differential equations carrying **gate variables** (ion-channel open probabilities), as in the Hodgkin–Huxley (HH) model.

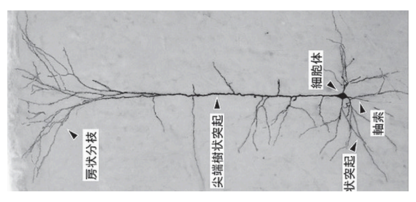
$$C_m \frac{dV}{dt} = -\bar{g}_L(V - E_L) - \bar{g}_{\mathrm{Na}} m^3 h (V - E_{\mathrm{Na}}) - \bar{g}_{\mathrm{K}} n^4 (V - E_{\mathrm{K}}) + I_{\mathrm{ext}}$$

HH membrane-potential equation (m, h, n: gate variables)

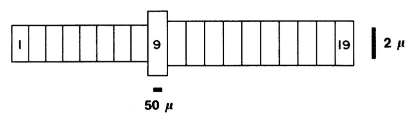
→ When many multi-compartment models are simulated in parallel (e.g. brain-scale simulation), the gate variables cause a **memory bottleneck**.

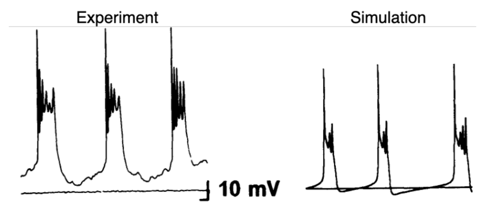
Goal

Build a surrogate model that reproduces the membrane-potential response of a multi-compartment neuron with **fewer state variables**.

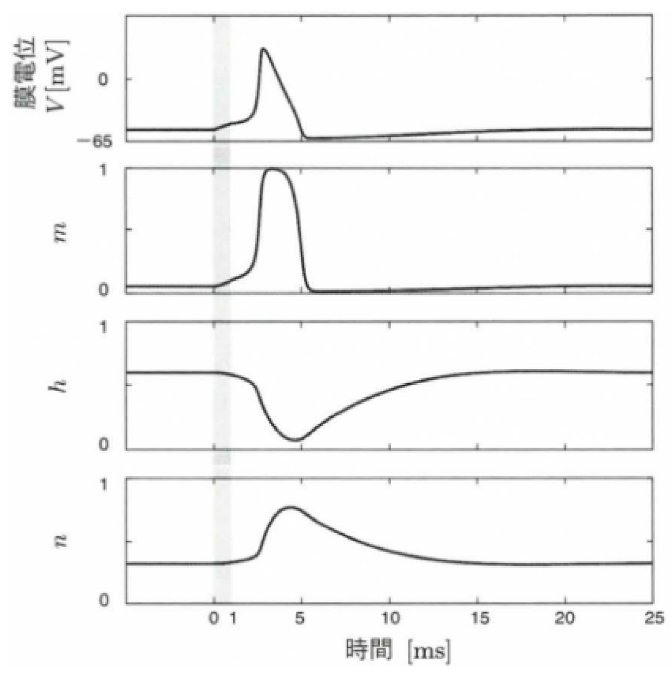


Pyramidal neuron [1]





Spatial representation and simulation of a multi-compartment model [2]



HH membrane-potential response [3]

Methods

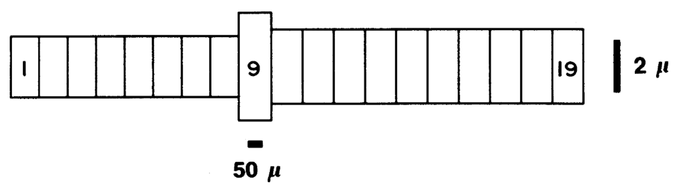
Two-axis surrogate framework

1. Simulate the target; collect V + gate time series.
2. **Compress** gates into a low-dim latent $z \in \mathbb{R}^n$ via a preprocessor — PCA (linear) or an **autoencoder** (nonlinear).
3. **Identify** the latent dynamics. The ansatz sets the form: **SINDy** picks from a generic library; **hybrid** keeps the ionic physics, fitting **only** the latent.

$$\begin{cases} \dfrac{dV}{dt} = \underbrace{I_{\mathrm{ion}}^{\mathrm{phys}}(V, \boldsymbol{z})}_{\text{known physics}} + I_{\mathrm{ext}} \\ \dfrac{d\boldsymbol{z}}{dt} = \underbrace{\Theta(\boldsymbol{z}, V)\boldsymbol{\xi}}_{\text{SINDy-identified}} \\ \boldsymbol{z} = \mathrm{AE}_{\mathrm{enc}}(\text{gates}), \quad \text{gates} = \mathrm{AE}_{\mathrm{dec}}(\boldsymbol{z}) \end{cases}$$

hybrid fixes last year's **over-large (41-term)** SINDy library.

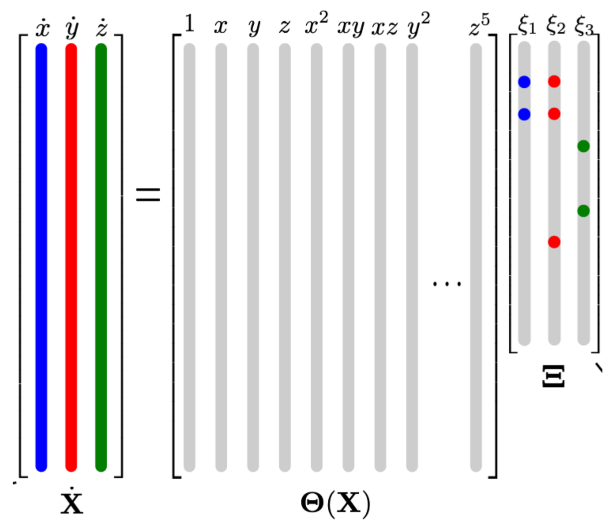
Target: Traub CA3 pyramidal cell — a 19-compartment realistic neuron (vs. last year's toy 3-comp).



Replace the soma with the surrogate: soma gates → $\boldsymbol{z}$; compartment coupling stays physical.

- preprocessor ∈ {PCA, AE}
- ansatz ∈ {SINDy, hybrid}
- latent dim $n = 2$ (HH) / 3 (Traub)

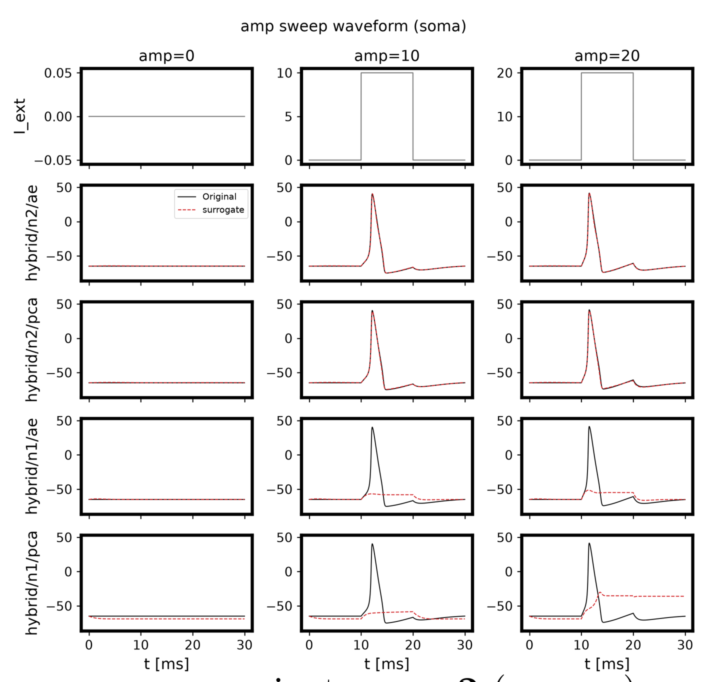
SINDy



Solve $\dot{X} = \Theta(X)\Xi$ (sparse regression) for Ξ [4]

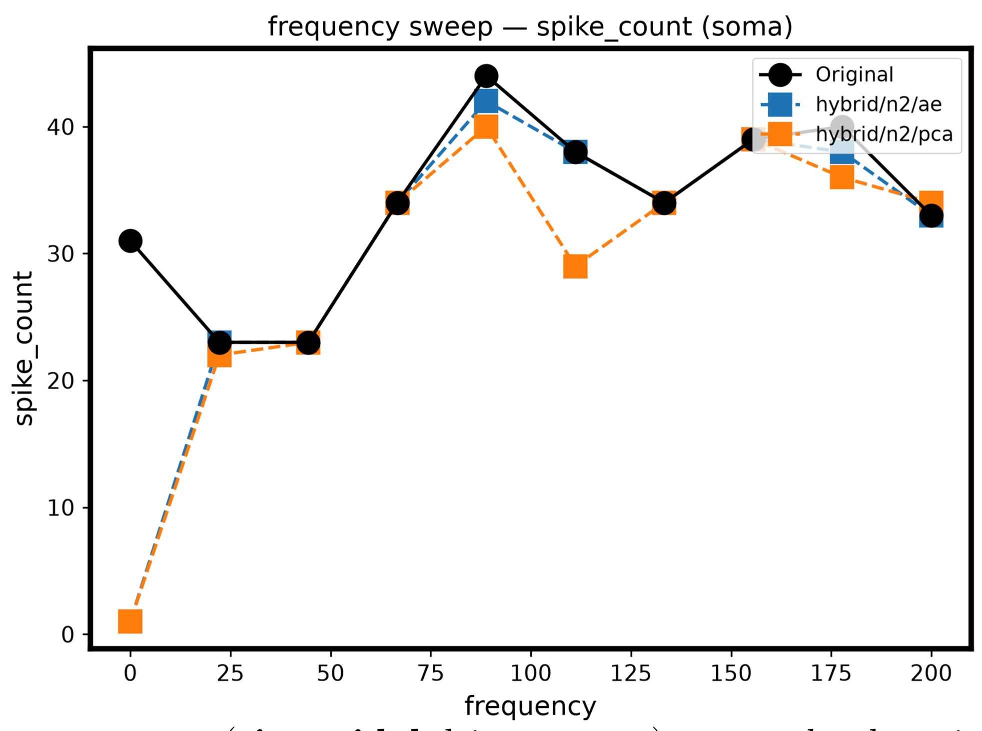
Results and Discussion

① Model selection — why hybrid/n2/ae?



Amplitude sweep across variants. $\boldsymbol{n = 2}$ (ae, pca) reproduces the spike; $\boldsymbol{n = 1}$ collapses.

- **Latent dim**: $n = 2$ is required — $n = 1$ cannot fire → drop $n = 1$.



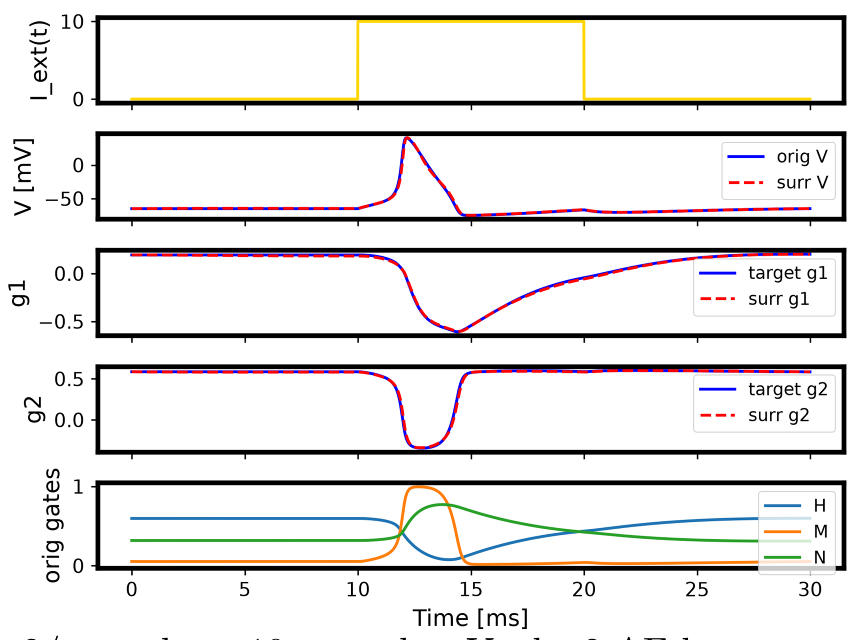
Frequency sweep (**sinusoidal** drive, unseen). **ae** tracks the original; **pca** mis-fires at $f = 0$ and dips near 110 Hz.

- **Preprocessor**: AE over PCA — the nonlinear encoder tracks 0–200 Hz → drop pca.

Selected

→ **hybrid / $\boldsymbol{n = 2}$ / AE**

② Single HH — accurate & interpretable

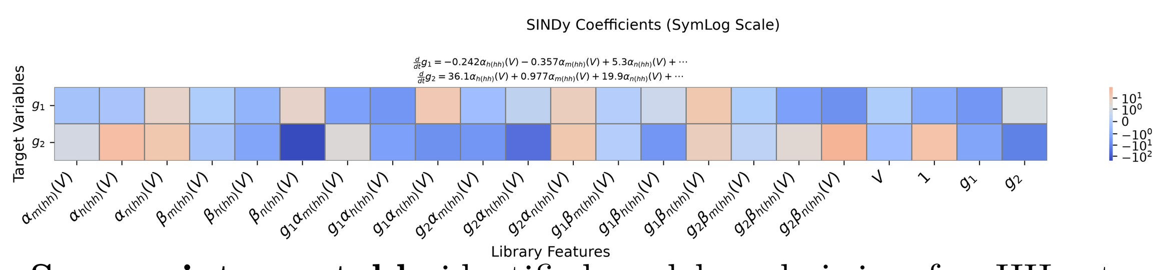


hybrid/n2/ae under a 10 ms pulse: V, the 2 AE latents g_1, g_2, and the 3 original gates. Original (blue) vs. surrogate (red, dashed) overlap.

Waveform match

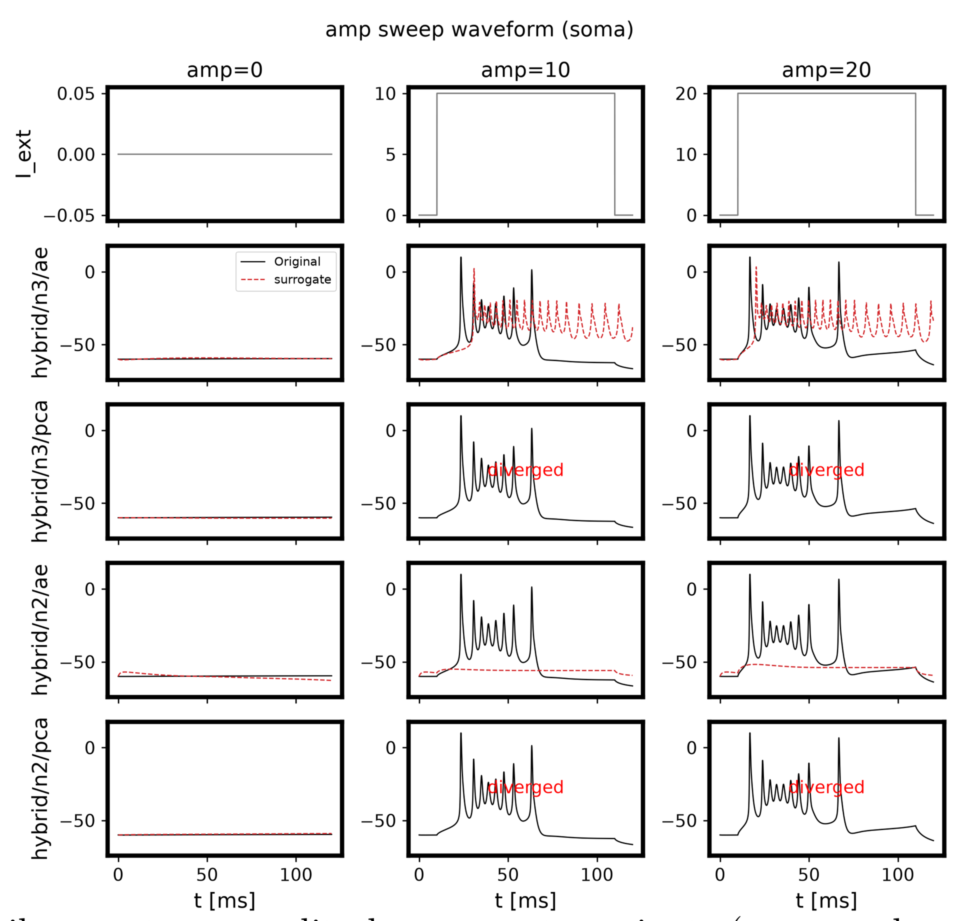
- RMSE **1.0 mV**, MAE **0.4 mV**, AP amp err **0.5 mV**
- spike count & latency **exact**

- **3 gates → 2 latents**, spike reproduced with **near-zero error**.



Sparse, interpretable identified model: each $\dot{g}_i$ is a few HH-rate-function terms.

③ Scaling up — Traub 19-compartment



Spike count vs. amplitude, surrogate variants (soma replaced).

- Same framework transfers **without diverging**; **only AE + $\boldsymbol{n = 3}$** fires (PCA, $n = 2$ collapse).
- **Gap**: spike count **overestimated** (9 vs. 6) — qualitative, not yet quantitative.

Conclusion

- HH: near-zero error at **dim 4 → 2**, generalizes to sinusoid, **sparse** latent eq.
- Scales to **Traub 19-comp** without diverging; qualitative firing recovered.

→ **Future**: close the spike-count gap, refine the physics-informed library.

Code — https://github.com/MunechikaHaruki/SINDyNeuroSurrogate

[1] "錐体細胞." Accessed: Dec. 08, 2025. [Online]. Available: https://doi.org/10.14931/bsd.2091

[2] R. D. Traub, R. K. Wong, R. Miles, and H. Michelson, "A Model of a CA3 Hippocampal Pyramidal Neuron Incorporating Voltage-Clamp Data on Intrinsic Conductances," *Journal of Neurophysiology*, vol. 66, no. 2, pp. 635–650, Aug. 1991, doi: 10.1152/jn.1991.66.2.635.

[3] 山﨑 匡 and 五十嵐 潤, はじめての神経回路シミュレーション: *1* ニューロンからヒト全脳モデルまで. 森北出版株式会社, 2021, pp. 56–61.

[4] K. Champion, B. Lusch, J. N. Kutz, and S. L. Brunton, "Data-Driven Discovery of Coordinates and Governing Equations," *Proceedings of the National Academy of Sciences*, vol. 116, no. 45, pp. 22445–22451, Nov. 2019, doi: 10.1073/pnas.1906995116.